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Basis Fields

A vector field may be written as a linear combination of basis vector fields. If n is the dimension, then any set of n linearly independent vector fields may be used as a basis. The coordinate basis $\mathbf{X}$ is an example of a basis.¹ We will see later that not every basis is a coordinate basis: in order to be a coordinate basis, there must be a coordinate system such that each basis element is the directional derivative operator in a corresponding coordinate direction.

Let $\mathbf{e}$ be a tuple of basis vector fields, such as the coordinate basis $\mathbf{X}$. The general vector field $\mathbf{v}$ applied to an arbitrary manifold function f can be expressed as a linear combination

$$\mathbf{v}(f)(\mathbf{m}) = \mathbf{e}(f)(\mathbf{m}) \cdot \mathbf{b}(\mathbf{m}) = \sum_i \mathbf{e}_i(f)(\mathbf{m}) b^i(\mathbf{m}), \quad (4.1)$$

where $\mathbf{b}$ is a tuple-valued coefficient function on the manifold. When expressed in a coordinate basis, the coefficients that specify the direction of the vector are naturally expressed as functions b^i of the coordinates of the manifold point. Here, the coefficient function $\mathbf{b}$ is more naturally expressed as a tuple-valued function on the manifold. If b is the coefficient function expressed as a function of coordinates, then $\mathbf{b} = b \circ \chi$ is the coefficient function as a function on the manifold.

The coordinate-basis forms have a simple definition in terms of the coordinate-basis vectors and the coordinates (equation 3.40). With this choice, the dual property, equation (3.41), holds without further fuss. More generally, we can define a basis of one-forms $\tilde{\mathbf{e}}$ that is dual to $\mathbf{e}$ in that the property

$$\tilde{\mathbf{e}}^i(\mathbf{e}_j)(\mathbf{m}) = \delta_j^i \quad (4.2)$$

is satisfied, analogous to property (3.41). Figure 4.1 illustrates the duality of basis fields.

¹We cannot say if the basis vectors are orthogonal or normalized until we introduce a metric.